Data Handling and Visualization

Lab Experiments

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Set 1

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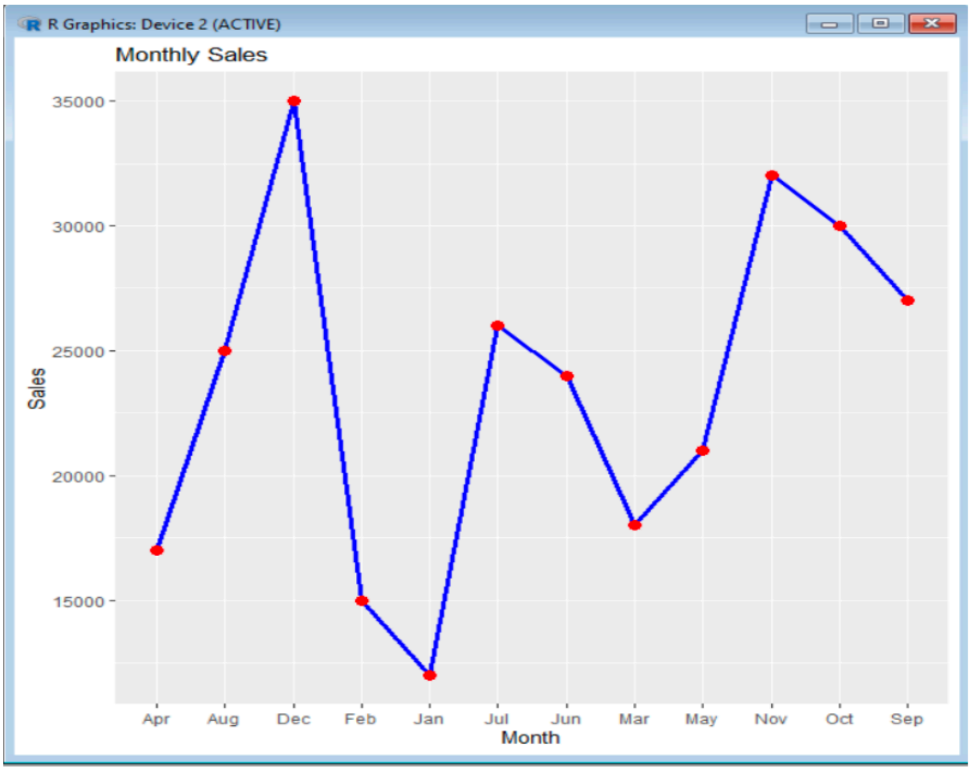
1. Line chart to visualize the monthly sales:
Python code:

```
sales <- data.frame(
Month = c("Jan","Feb","Mar","Apr","May","Jun",
"Jul","Aug","Sep","Oct","Nov","Dec"),
Sales = c(12000,15000,18000,17000,21000,24000,
26000,25000,27000,30000,32000,35000),
Product = c("A","B","C","D","E","F",
"G","H","I","J","K","L"),
AdBudget = c(2000,2500,3000,2800,3500,4000,
4200,4500,4800,5000,5500,6000)
)
ggplot(sales, aes(x = Month, y = Sales, group = 1)) +
geom_line(color = "blue", size = 1.2) +
geom_point(size = 3, color = "red") +
labs(title = "Monthly Sales",
x = "Month",
y = "Sales")
```

Output:

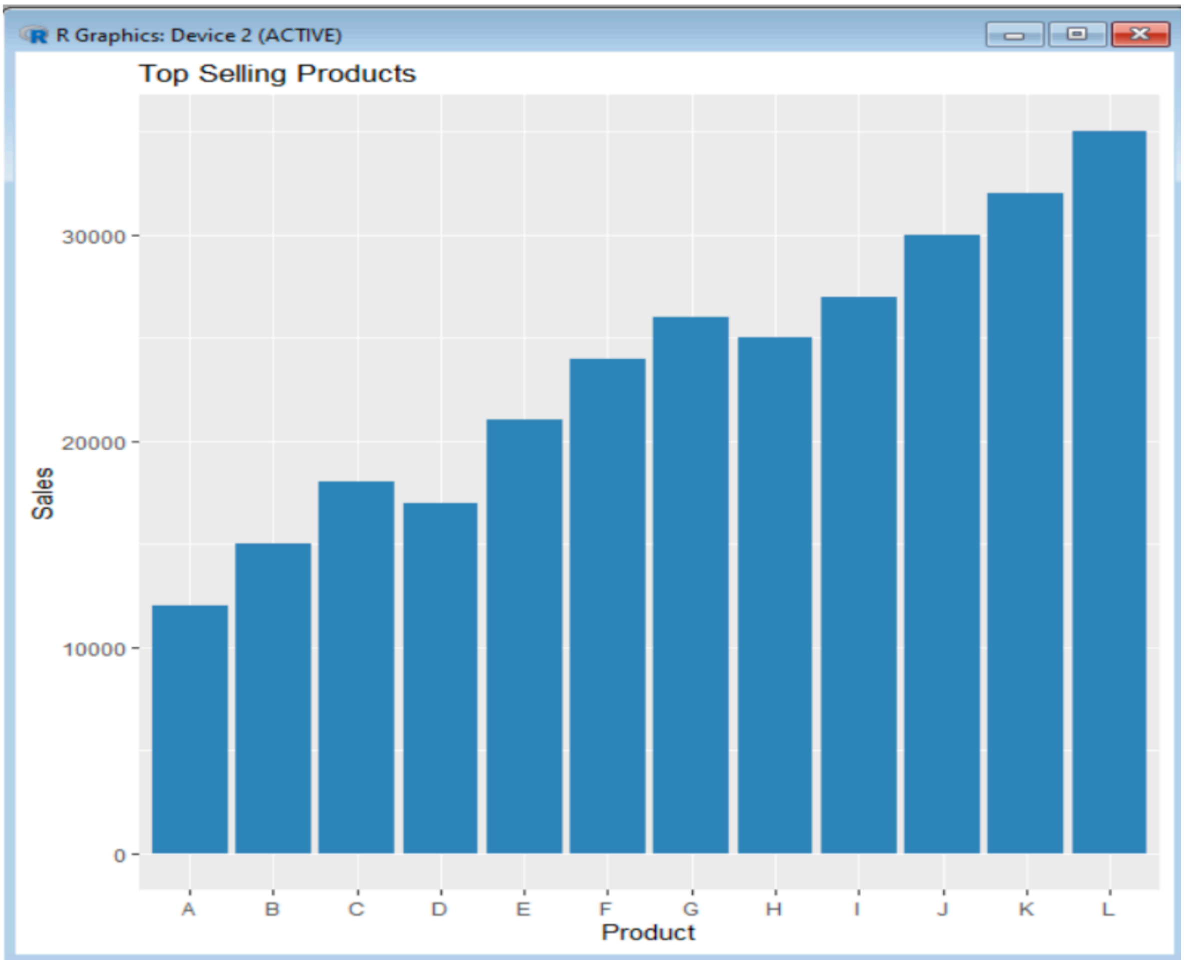


2. Top-selling products for the year:

Python code:

```
sales <- data.frame(
Month = c("Jan","Feb","Mar","Apr","May","Jun",
"Jul","Aug","Sep","Oct","Nov","Dec"),
Sales = c(12000,15000,18000,17000,21000,24000,
26000,25000,27000,30000,32000,35000),
Product = c("A","B","C","D","E","F",
"G","H","I","J","K","L"),
AdBudget = c(2000,2500,3000,2800,3500,4000,
4200,4500,4800,5000,5500,6000)
)
ggplot(sales, aes(x = Product, y = Sales)) +
geom_bar(stat = "identity", fill = "steelblue") +
labs(title = "Top Selling Products",
x = "Product",
y = "Sales")
```

Output:

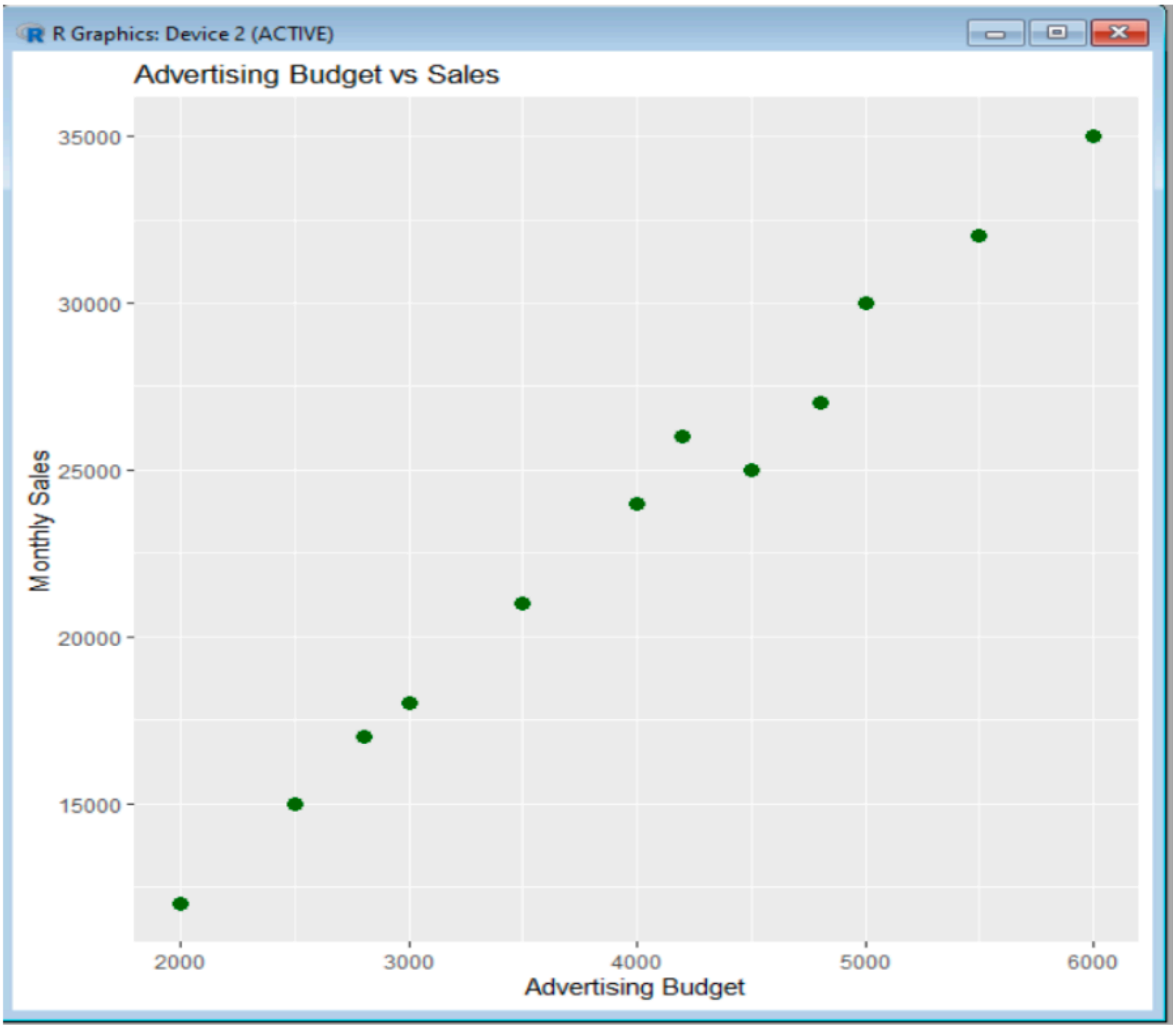


3. The relationship between advertising budget and monthly sales:

Python code:

```
sales <- data.frame(
Month = c("Jan","Feb","Mar","Apr","May","Jun",
"Jul","Aug","Sep","Oct","Nov","Dec"),
Sales = c(12000,15000,18000,17000,21000,24000,
26000,25000,27000,30000,32000,35000),
Product = c("A","B","C","D","E","F",
"G","H","I","J","K","L"),
AdBudget = c(2000,2500,3000,2800,3500,4000,
4200,4500,4800,5000,5500,6000)
)
ggplot(sales, aes(x = AdBudget, y = Sales)) +
geom_point(size = 3, color = "darkgreen") +
labs(title = "Advertising Budget vs Sales",
x = "Advertising Budget",
y = "Monthly Sales")
```

Output:



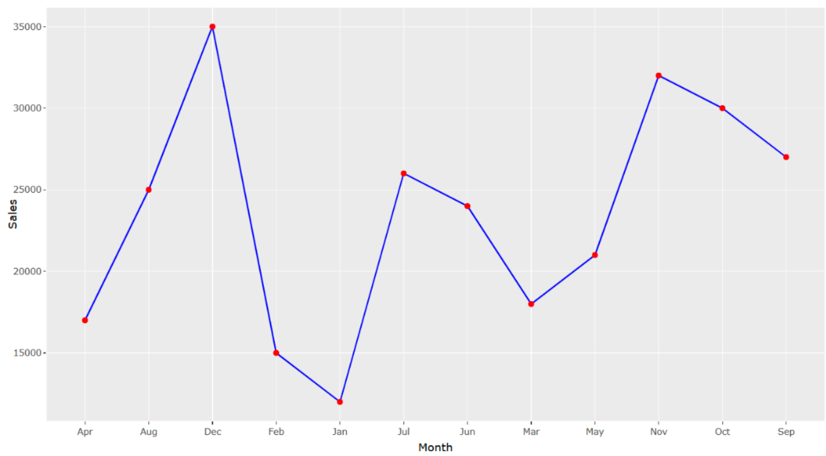
4. Combining the line chart and bar chart:

4a. Line chart:

Python code:

```
sales <- data.frame(
Month = c("Jan","Feb","Mar","Apr","May","Jun",
"Jul","Aug","Sep","Oct","Nov","Dec"),
Sales = c(12000,15000,18000,17000,21000,24000,
26000,25000,27000,30000,32000,35000))
p1 <- ggplot(sales, aes(x = Month, y = Sales, group = 1)) +
geom_line(color = "blue") +
geom_point(color = "red")
ggplotly(p1)
```

Output:

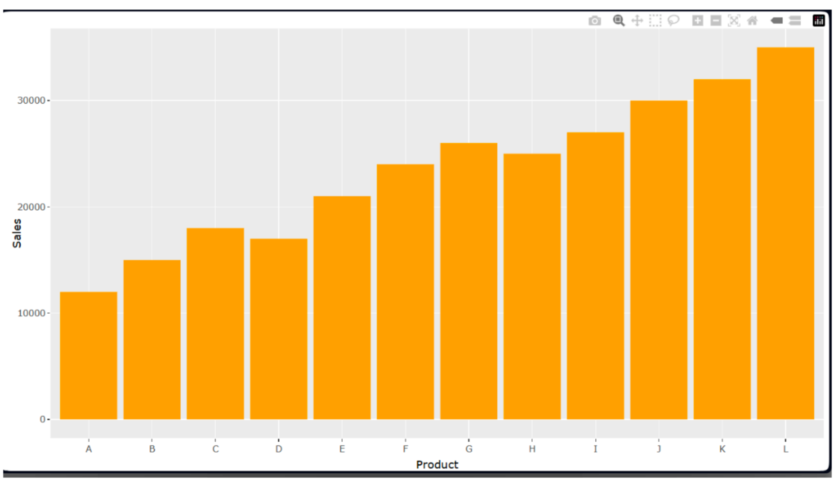


4b. bar chart:

Python code:

```
sales <- data.frame(
Product = c("A","B","C","D","E","F","G","H","I","J","K","L"),
Sales = c(12000,15000,18000,17000,21000,24000,
26000,25000,27000,30000,32000,35000)
)
p2 <- ggplot(sales, aes(x = Product, y = Sales)) +
geom_bar(stat = "identity", fill = "orange")
ggplotly(p2)
```

Output:



Set 2

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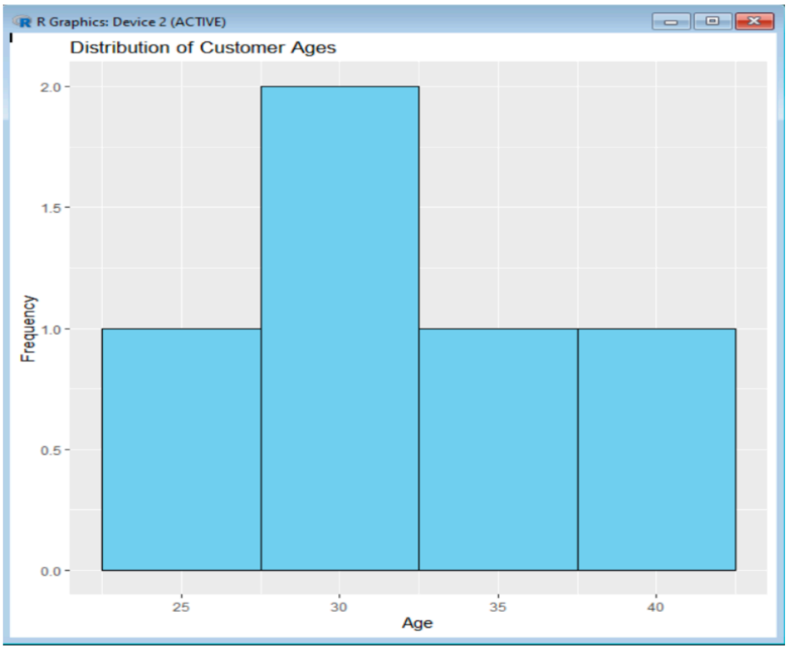
1. Histogram to represent the distribution of customer ages:

Python code:

```
customer <- data.frame(
CustomerID = c(1,2,3,4,5),
Age = c(25,30,35,28,40),
Satisfaction = c(4,5,3,4,5)
)
ggplot(customer, aes(x = Age)) +
geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
labs(title = "Distribution of Customer Ages",
x = "Age",
y = "Frequency")
```

Output:



2. Pie chart to display the overall distribution of customer satisfaction scores:

Python code:

```
customer <- data.frame(
CustomerID = c(1,2,3,4,5),
Age = c(25,30,35,28,40),
Satisfaction = c(4,5,3,4,5)
)
pie_data <- table(customer$Satisfaction)
pie(
pie_data,
labels = names(pie_data),
col = rainbow(length(pie_data)),
main = "Customer Satisfaction Scores")
```

Output:

3. Bar chart to visualize the distribution of customer satisfaction scores by age group:

Python code:

```
customer$AgeGroup <- cut(
customer$Age,
breaks = c(20,30,40,50),
labels = c("21-30","31-40","41-50"),
include.lowest = TRUE)
ggplot(customer,
aes(x = AgeGroup,
fill = factor(Satisfaction))) +
geom_bar() +
labs(title = "Customer Satisfaction by Age Group",
x = "Age Group",y = "Count",
fill = "Satisfaction")
```

Output:

4. Word cloud from open-ended customer feedback to identify prevalent customet sentiments:

Python code:

```
feedback <- c(
"Good service",
"Excellent support",
"Friendly staff",
"Fast delivery",
"Very satisfied",
"Helpful service",
"Great experience",
"Quick response",
"Quality product",
"Happy customer"
)
wordcloud(
words = feedback,
random.order = FALSE,
colors = brewer.pal(8, "Dark2")
)
```

Output:

Set 3

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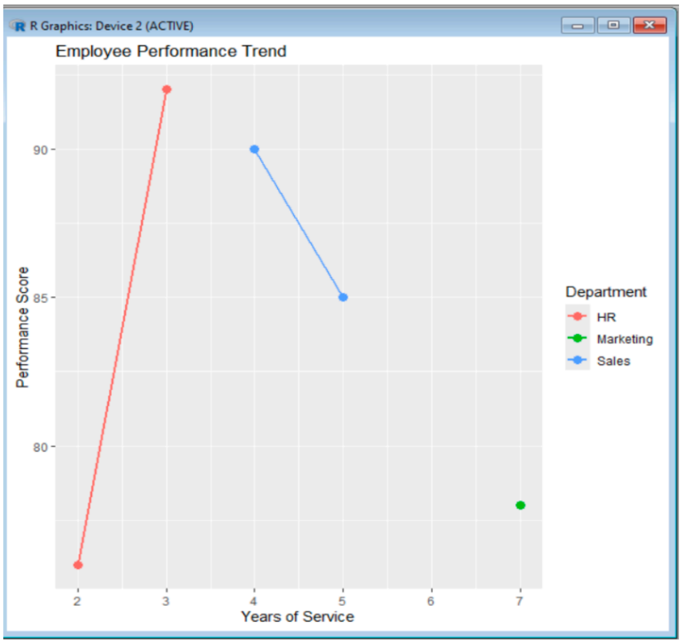
1. Line chart to visualize the performance trend of employees over time:

Python code:

```
employee <- data.frame(
EmployeeID = c(1,2,3,4,5),
Department = c("Sales","HR","Marketing","Sales","HR"),
YearsService = c(5,3,7,4,2),
Performance = c(85,92,78,90,76)
```

```
)
ggplot(employee,
aes(x = YearsService,y = Performance,
color = Department,
group = Department)) +
geom_line(size = 1) +
geom_point(size = 3) +
labs(title = "Employee Performance Trend",
x = "Years of Service",
y = "Performance Score",
color = "Department")
```

Output:

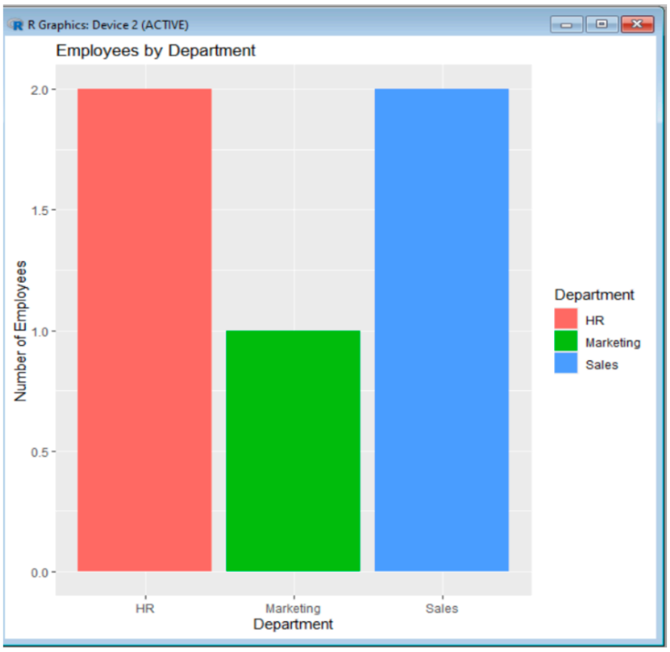


2. Bar chart showing the distribution of employees across different departments:

Python code:

```
employee <- data.frame(
EmployeeID = c(1,2,3,4,5),
Department = c("Sales","HR","Marketing","Sales","HR"),
YearsService = c(5,3,7,4,2),
Performance = c(85,92,78,90,76))
ggplot(employee,
aes(x = Department,
fill = Department)) +
geom_bar() +
labs(title = "Employees by Department",
x = "Department",
y = "Number of Employees")
```

Output:

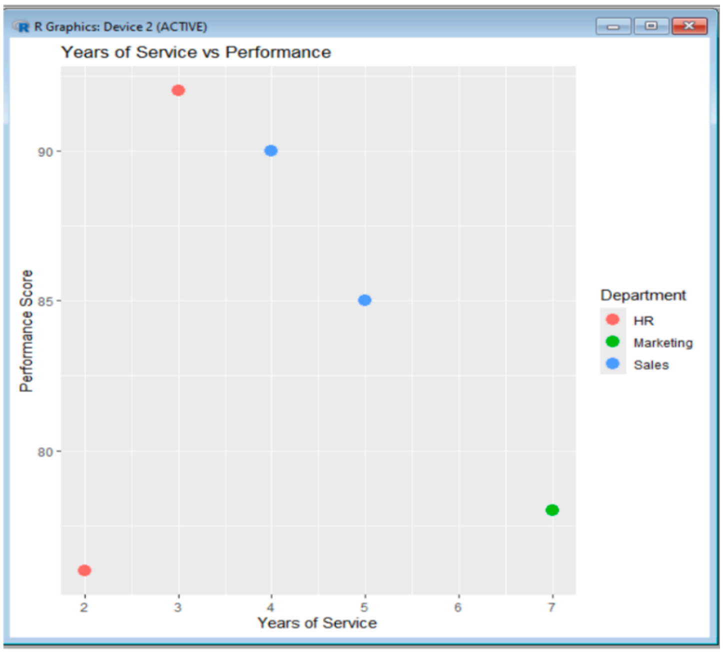


3. Scatter plot to analyze the correlation between years of service and performance scores:

Python code:

```
employee <- data.frame(
EmployeeID = c(1,2,3,4,5),
Department = c("Sales","HR","Marketing","Sales","HR"),
YearsService = c(5,3,7,4,2),
Performance = c(85,92,78,90,76)
)
ggplot(employee,aes(x = YearsService,
y = Performance,
color = Department)) +
geom_point(size = 4) +
labs(title = "Years of Service vs Performance",
x = "Years of Service",
y = "Performance Score")
```

Output:



4. Tableau dashboard with the line chart and bar chart for interactive exploration:
Output:

Set 4

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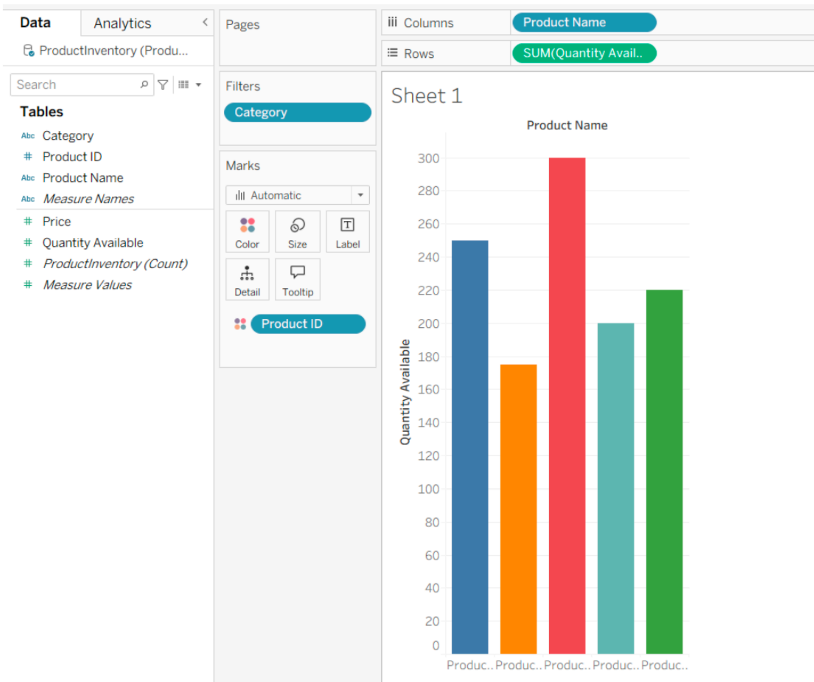
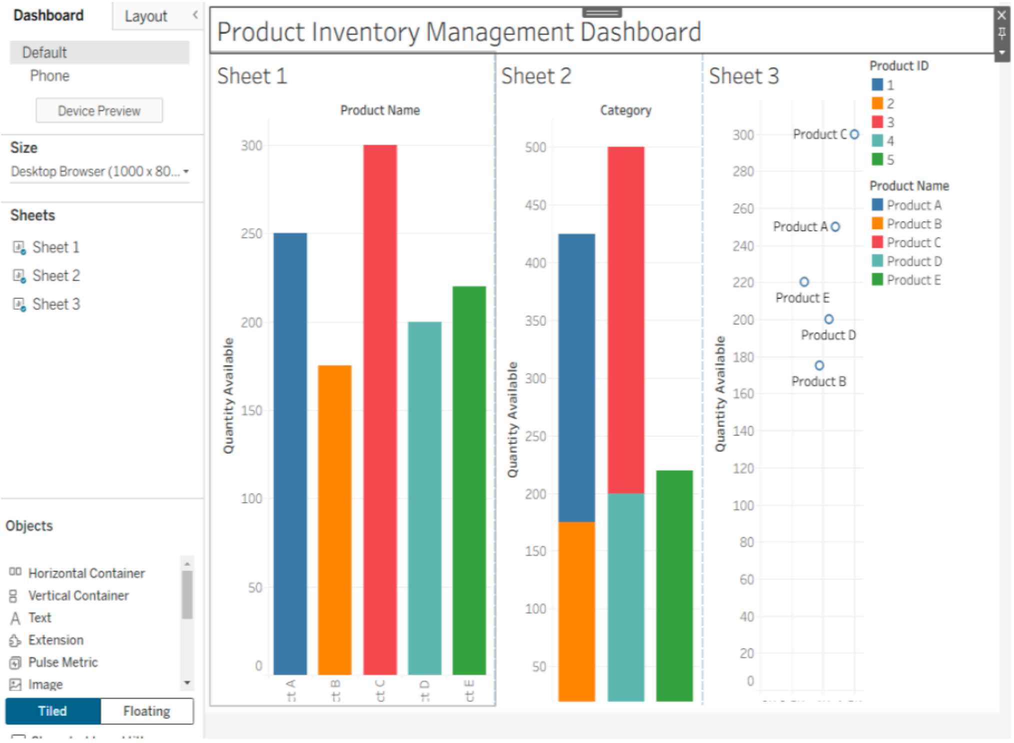
1. Bar chart to visualize the quantity of each product in the inventory:
Output:



2. stacked bar chart to show the quantity of each product within different product categories:
Output:

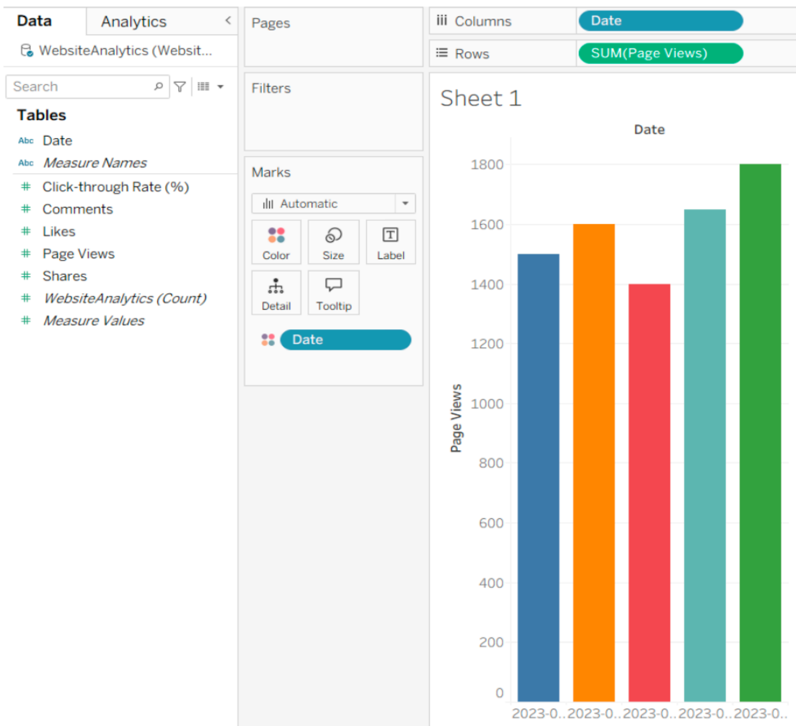
3. scatter plot to explore the relationship between product price and quantity available:
Output:

4. Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.
Output:

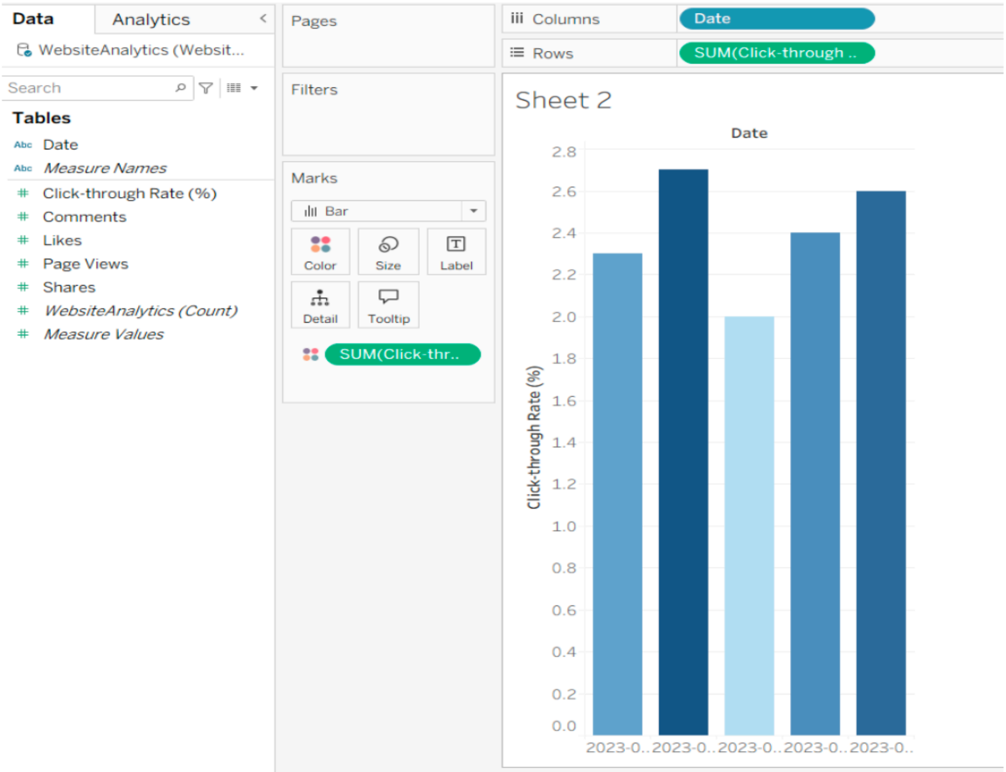


Set 5

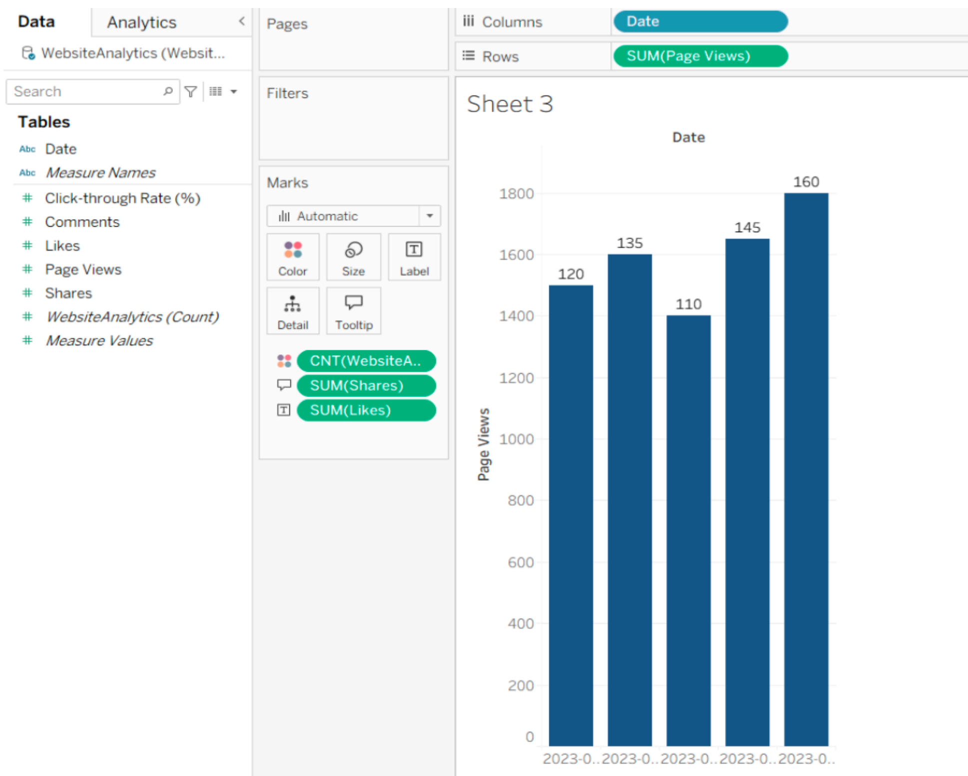
1. line chart to visualize the trend in daily page views over time:
Output:



2. bar chart showing the top N days with the highest click-through rates:
Output:



3. stacked area chart to display the distribution of user interactions:
Output:



4. Tableau dashboard with the line chart and bar chart for interactive exploration of website:
Output: